

OPTICS MCQ

1. Optics branch concerned with light behavioural pattern properties

- a) sound waves
- b) electromagnetic radiation
- c) mechanical waves
- d) thermal energy

Correct answer: b

2. Physical optics uses wave nature represents light

- a) straight rays
- b) transverse wavefront
- c) photons only
- d) particles

Correct answer: b

3. Geometrical optics treats light

- a) wave interference
- b) straight line rays
- c) quantum packets
- d) diffraction patterns

Correct answer: b

4. Quantum optics studies individual light quanta called

- a) waves
- b) photons
- c) electrons
- d) protons

Correct answer: b

5. Light produces sensation produces

- a) hearing
- b) sight
- c) touch
- d) smell

Correct answer: b

6. Luminous objects emit

- a) absorb light
- b) light energy
- c) reflect only
- d) transmit only

Correct answer: b

7. Speed light vacuum equals

- a) 3×10^6 m/s
- b) 3×10^8 m/s
- c) 3×10^{10} m/s
- d) 3×10^4 m/s

Correct answer: b

8. Light frequency remains constant across

- a) wavelengths change
- b) all media
- c) vacuum only
- d) air only

Correct answer: b

9. Wave equation relates c equals

- a) wavelength frequency

- b) speed amplitude
- c) energy momentum
- d) phase velocity

Correct answer: a

10. Violet light possesses lowest

- a) frequency
- b) wavelength
- c) speed
- d) energy

Correct answer: b

11. Reflection occurs polished

- a) rough surfaces
- b) metal mirrors
- c) paper
- d) cloth

Correct answer: b

12. Refraction caused difference light speeds

- a) same media
- b) different media
- c) vacuum air
- d) glass water

Correct answer: b

13. First refraction law incident refracted normal same

- a) curve
- b) plane
- c) focus

d) axis

Correct answer: b

14. Snell's law ratio sine angles equals refractive

a) indices ratio

b) speeds ratio

c) densities ratio

d) wavelengths ratio

Correct answer: a

15. Refractive index defined speed vacuum divided speed

a) medium

b) air

c) water

d) glass

Correct answer: a

16. Rarer medium light bends

a) towards normal

b) away normal

c) parallel normal

d) perpendicular normal

Correct answer: b

17. Dispersion splits white light constituent

a) three colours

b) seven colours

c) two colours

d) infinite colours

Correct answer: b

18.VIBGYOR spectrum violet deviated

- a) least
- b) most
- c) equal
- d) none

Correct answer: b

19.Rainbow formation example

- a) reflection
- b) dispersion
- c) scattering
- d) interference

Correct answer: b

20.Scattering light absorbed re-radiated

- a) same direction
- b) random directions
- c) opposite direction
- d) forward only

Correct answer: b

21.Elastic scattering incident scattered energies

- a) different
- b) same
- c) increased
- d) decreased

Correct answer: b

22.Rayleigh scattering inversely proportional wavelength

- a) first power

b) second power

c) third power

d) fourth power

Correct answer: d

23. Blue sky appears due greater scattering

a) red light

b) shorter wavelengths

c) longer wavelengths

d) green light

Correct answer: b

24. Sunrise sunset red colour least

a) absorption

b) scattering

c) refraction

d) reflection

Correct answer: b

25. Mie scattering particle size comparable

a) wavelength

b) frequency

c) speed

d) energy

Correct answer: a

26. Clouds appear white equal scattering

a) all colours

b) blue only

c) red only

d) violet only

Correct answer: a

27. Tyndall effect observed

a) true solutions

b) colloids suspensions

c) crystals

d) gases

Correct answer: b

28. Raman scattering scattered light contains additional

a) amplitudes

b) frequencies

c) speeds

d) phases

Correct answer: b

29. Convex lens thicker

a) edges

b) centre

c) both equal

d) neither

Correct answer: b

30. Concave lens called diverging because rays

a) converge

b) spread out

c) parallel

d) reflect

Correct answer: b

31. Plano-convex lens one surface

- a) concave
- b) plane
- c) convex both
- d) cylindrical

Correct answer: b

32. Ray through optical centre lens

- a) bends
- b) undeviated
- c) reflects
- d) absorbs

Correct answer: b

33. Parallel rays convex lens converge

- a) optical centre
- b) principal focus
- c) infinity
- d) centre curvature

Correct answer: b

34. Object infinity convex lens image

- a) infinity
- b) focus
- c) $2F$
- d) optical centre

Correct answer: b

35. Object centre curvature convex image same size

- a) inverted real

b) erect virtual

c) diminished

d) enlarged

Correct answer: a

36. Object focus convex image formed

a) focus

b) $2F$

c) infinity

d) optical centre

Correct answer: c

37. Virtual image convex lens object between focus

a) optical centre

b) $2F$

c) infinity

d) C

Correct answer: a

38. Concave lens object infinity image

a) real

b) virtual focus

c) enlarged

d) inverted

Correct answer: b

39. Hypermetropia corrected

a) concave lens

b) convex lens

c) cylindrical lens

d) bifocal lens

Correct answer: b

40. Lens formula relates object distance image distance

a) wavelength

b) focal length

c) power

d) magnification

Correct answer: b

41. Sign convention object placed lens

a) right side

b) left side

c) above axis

d) below axis

Correct answer: b

42. Magnification height image divided height

a) object

b) lens

c) focus

d) aperture

Correct answer: a

43. Lens maker formula relates focal length refractive index

a) diameters

b) radii curvature

c) thicknesses

d) apertures

Correct answer: b

44. Power lens equals reciprocal

- a) diameter
- b) focal length
- c) refractive index
- d) magnification

Correct answer: b

45. Diopetre SI unit lens

- a) power
- b) focal length
- c) magnification
- d) aperture

Correct answer: a

46. Human eye diameter approximately

- a) 1 cm
- b) 2.3 cm
- c) 5 cm
- d) 10 cm

Correct answer: b

47. Cornea functions light

- a) absorbs
- b) bends enters
- c) reflects
- d) scatters

Correct answer: b

48. Iris controls light entering through

- a) sclera

b) pupil

c) retina

d) lens

Correct answer: b

49. Retina contains light-sensitive

a) muscles

b) nerve cells

c) fluids

d) blood vessels

Correct answer: b

50. Cones sensitive

a) dim light

b) bright colour vision

c) peripheral vision

d) motion only

Correct answer: b

51. Power accommodation adjusts focal length focus

a) near far objects

b) same distance

c) infinity only

d) 25 cm only

Correct answer: a

52. Ciliary muscles alter lens

a) colour

b) curvature

c) transparency

d) size

Correct answer: b

53. Persistence vision retinal impression lasts

a) 1 second

b) 1/16 second

c) 1 minute

d) 1 hour

Correct answer: b

54. Normal eye near point

a) 10 cm

b) 25 cm

c) 50 cm

d) infinity

Correct answer: b

55. Myopia image forms

a) behind retina

b) retina

c) before retina

d) infinity

Correct answer: c

56. Myopia corrected placing

a) convex lens

b) concave lens

c) bifocal

d) cylindrical

Correct answer: b

57. Hypermetropia near point moves

- a) closer
- b) farther
- c) infinity
- d) focus

Correct answer: b

58. Presbyopia due ageing ciliary muscles

- a) strengthen
- b) weaken
- c) enlarge
- d) shrink

Correct answer: b

59. Bifocal lens upper part corrects

- a) hypermetropia
- b) myopia
- c) presbyopia
- d) astigmatism

Correct answer: b

60. Astigmatism corrected

- a) spherical lenses
- b) cylindrical lenses
- c) convex lenses
- d) concave lenses

Correct answer: b

61. Simple microscope single

- a) concave lens

- b) convex short focal length
- c) compound lens
- d) mirror

Correct answer: b

62. Compound microscope two lenses objective

- a) eye lens
- b) power lens
- c) plane lens
- d) cylindrical lens

Correct answer: a

63. Objective lens shorter

- a) aperture
- b) focal length
- c) diameter
- d) power

Correct answer: b

64. Travelling microscope measures small lengths accuracy

- a) 0.1 mm
- b) 0.01 mm
- c) 1 mm
- d) 0.001 mm

Correct answer: b

65. Telescope views

- a) nearby objects
- b) distant objects
- c) microscopic objects

d) coloured objects

Correct answer: b

66. First telescope invented

a) Galileo

b) Lippershey

c) Kepler

d) Newton

Correct answer: b

67. Reflecting telescope uses

a) convex lenses

b) parabolic mirrors

c) concave lenses

d) prisms

Correct answer: b

68. Astronomical telescope views

a) earthly objects

b) heavenly bodies

c) insects

d) cells

Correct answer: b

69. Terrestrial telescope produces

a) inverted image

b) erect image

c) virtual image

d) diminished image

Correct answer: b

70. Refracting telescope disadvantage chromatic

- a) aberration
- b) distortion
- c) magnification
- d) resolution

Correct answer: a

71. Light transverse wave requires

- a) medium travel
- b) no medium vacuum
- c) solid medium
- d) liquid medium

Correct answer: b

72. Angle incidence equals angle

- a) refraction
- b) reflection
- c) deviation
- d) dispersion

Correct answer: b

73. Velocity higher

- a) denser medium
- b) rarer medium
- c) same all media
- d) vacuum only

Correct answer: b

74. Red light deviated prism

- a) most

- b) least
- c) equal violet
- d) absorbed

Correct answer: b

75. Stokes lines Raman scattering frequency

- a) higher incident
- b) lower incident
- c) equal incident
- d) double incident

Correct answer: b

76. Anti-stokes lines frequency

- a) lower
- b) higher incident
- c) equal
- d) half

Correct answer: b

77. Object beyond $2F$ convex image between

- a) F $2F$
- b) infinity F
- c) O F
- d) C infinity

Correct answer: a

78. Concave lens always produces

- a) real images
- b) virtual images
- c) inverted images

d) enlarged images

Correct answer: b

79. Focal length positive

a) concave lens

b) convex lens

c) plane glass

d) prism

Correct answer: b

80. Object distance lens formula always

a) positive

b) negative

c) zero

d) infinity

Correct answer: b

81. Magnification positive indicates image

a) inverted

b) erect

c) real

d) diminished

Correct answer: b

82. Eye lens becomes thinner viewing

a) near objects

b) distant objects

c) both

d) neither

Correct answer: b

83. Blind spot lacks

- a) cones
- b) rods
- c) sensory cells
- d) iris

Correct answer: c

84. Myopic eye far point finite distance

- a) infinity
- b) 25 cm
- c) x cm
- d) 50 cm

Correct answer: c

85. Hypermetropic near point

- a) 25 cm
- b) $d > 25 \text{ cm}$
- c) infinity
- d) 10 cm

Correct answer: b

86. Simple microscope magnification maximum least distance

- a) distinct vision
- b) infinity
- c) focus
- d) $2F$

Correct answer: a

87. Compound microscope total magnification objective times

- a) eyepiece

b) single lens

c) prism

d) mirror

Correct answer: a

88. Galilean telescope uses

a) two convex

b) convex objective concave eyepiece

c) two concave

d) mirrors

Correct answer: b

89. Newtonian telescope

a) refracting

b) reflecting

c) terrestrial

d) astronomical

Correct answer: b

90. Telescope advantage low light

a) poor performance

b) good performance

c) equal performance

d) no vision

Correct answer: b

91. Dispersion proportional inversely

a) wavelength

b) frequency

c) speed

d) amplitude

Correct answer: a

92. Rayleigh scattering explains

a) rainbow

b) blue sky

c) white clouds

d) red sunset

Correct answer: b

93. Mie scattering wavelength

a) dependent

b) independent

c) proportional

d) inversely proportional

Correct answer: b

94. Power accommodation decreases

a) youth

b) ageing

c) exercise

d) sleep

Correct answer: b

95. Bifocal lens invented

a) Galileo

b) Benjamin Franklin

c) Newton

d) Kepler

Correct answer: b

96.Travelling microscope vernier scale least count

- a) 0.1 mm
- b) 0.01 mm
- c) 1 mm
- d) 0.001 mm

Correct answer: b

97.Kepler telescope

- a) erect image
- b) inverted image
- c) virtual image
- d) real image

Correct answer: b

98.Cassegrain telescope

- a) refracting
- b) reflecting
- c) compound
- d) simple

Correct answer: b

99.Lens power increases decreasing

- a) diameter
- b) focal length
- c) thickness
- d) refractive index

Correct answer: b

100. Normal vision range 25 cm to

- a) 50 cm

b) 100 cm

c) infinity

d) 10 m

Correct answer: c